

Research Interests

Fairness, Pricing, Transportation Systems, Machine Learning

Education

- Aug. 2021 – **Ph.D. in Operations Research**, *Columbia University*, New York, USA
present Advisor: Prof. Adam N. Elmachtoub
- Sept. 2018 – **M.S. in Industrial & Systems Engineering**, *KAIST*, Daejeon, Korea
Aug. 2020 Advisor: Prof. Il-Chul Moon
- Feb. 2012 – **B.S. in Industrial & Systems Engineering and Electrical Engineering**, *KAIST*, Daejeon, Korea
Aug. 2018 Summa Cum Laude

Publication

- [1] **Fair Fares for Vehicle Sharing Systems**, Adam N. Elmachtoub and Hyemi Kim
Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency, 2025.
○ Finalist, *INFORMS DEI Best Student Paper Award*, 2024
○ Finalist, *INFORMS Transportation Science and Logistics Society Best Student Paper Award*, 2024
- [2] **Black-Box EM Algorithm for Estimating Latent States of High-Speed Vehicles**, Yoon-Yeong Kim, Hyemi Kim, WonSung Lee, Han-Lim Choi, and Il-Chul Moon
American Institute of Aeronautics and Astronautics, 2021.
- [3] **Counterfactual Fairness with Disentangled Causal Effect Variational Autoencoder**, Hyemi Kim, Seungjae Shin, JoonHo Jang, Kyungwoo Song, Weonyoung Joo, Wanmo Kang, and Il-Chul Moon
Association for the Advancement of Artificial Intelligence (AAAI), 2021.
- [4] **Neutralizing Gender Bias in Word Embedding with Latent Disentanglement and Counterfactual Generation**, Seungjae Shin, Kyungwoo Song, Joonho Jang, Hyemi Kim, Weonyoung Joo, and Il-Chul Moon
Findings of Empirical Methods in Natural Language Processing (Findings of EMNLP), 2020.
- [5] **Deep Generative Positive-Unlabeled Learning under Selection Bias**, ByeongHu Na, Hyemi Kim, Kyungwoo Song, Weonyoung Joo, Yoonyeong Kim, and Il-Chul Moon
Conference on Information and Knowledge Management (CIKM), 2020.

Honors and Awards

- 2025 Columbia-Dream Sports AI Innovation Fellowship, Columbia University
- 2024 Finalist, *INFORMS DEI Best Student Paper Award*
- 2024 Finalist, *INFORMS Transportation Science and Logistics Best Student Paper Award*
- 2023 Deming Doctoral Fellowship, Columbia University
- 2019 Songhyun Awards, KAIST
- 2012 Presidential Science Scholarship, KFAS

Talks

Learning Fair Demand Models

- 2025 INFORMS Annual Meeting, Atlanta, Georgia
- 2025 INFORMS RMP Conference, New York, NY

Fair Fares for Vehicle Sharing Systems

- 2025 ACM FAccT Conference, Athena, Greek
- 2025 INFORMS TSL Workshop, Seoul, Korea
- 2024 INFORMS DEI & TSL Best Student Paper Awards, Seattle, WA
- 2024 INFORMS RMP Conference, Los Angeles, CA
- 2024 Market Innovation Workshop, Virtual
- 2023 INFORMS Annual Meeting, Phoenix, AZ
- 2023 Workshop on AI & Analytics for Social Good, College Park, MD

Teaching Assistant

- Fall 2024 **Introduction to Probability and Statistics**, Columbia IEOR
- Spring 2024 **Stochastic Models**, Columbia IEOR
- Fall 2023 **Analytics in Action**, Columbia Business School
- Spring 2023 **Transportation Analytics & Logistics**, Columbia IEOR
- Spring 2022 **Causal Inference**, Columbia IEOR
- Fall 2019 **Data Structure**, KAIST ISysE
- Fall 2018 **Data Structure**, KAIST ISysE

Academic Service

- 2025 Program Committee, Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO)
- 2025 Program Committee, IJCAI – Special Track on AI and Social Good
- 2024 Program Committee, Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO)

Industry Experience

- Jun. 2024 – **Data Scientist Intern**, *MOLOCO, Inc.*, Redwood City, USA
- Aug. 2024 Designed and executed budget pacing simulations for online advertisements.
- Jan. 2021 – **AI Research Intern**, *NAVER Corporation*, Seongnam, Korea
- Jul. 2021 Studied test-time training methods for robust machine learning.
- Jun. 2015 – **Intern**, *Samsung Electronics*, Suwon, Korea
- Aug. 2015 Analyzed smart car development trends and improved patient analysis algorithms.

Skills & Personal

- Programming Python, MATLAB, C
- Frameworks PyTorch, TensorFlow, Gurobi, Pyomo
- Languages English (advanced), Korean (native)
- Hobby Brazilian jiu-jitsu